

(TR-104) PYTHON WITH AI/ML

Training Day 74 Report

Date: 23 April 2026

Topic Covered: MongoDB Analysis, FastAPI Development, AI Agent Integration & Performance Optimization

The seventy-fourth day of training focused on analyzing MongoDB data, developing FastAPI-based backend APIs, implementing caching mechanisms, integrating AI agents using Groq and Ollama, and optimizing the AI curriculum structure.

- Analyzed the MB Trailers MongoDB database and identified data inconsistency issues across collections.
- Created a sample MongoDB database (leasedb) with trailers, customers, and lease payments collections using ObjectId relationships.
- Developed FastAPI backend APIs and integrated MongoDB using the Motor asynchronous driver.
- Implemented reusable MongoDB aggregation pipelines using \$lookup, \$unwind, and \$project operators.
- Developed APIs for payment tracking, customer details, dashboard summaries, and trailer filtering.
- Implemented in-memory caching and optimized database queries to improve performance.
- Integrated AI agents using Ollama and Groq with tool-based querying and structured response generation.
- Enhanced AI curriculum structure by refining LangChain, RAG, Agents, and deployment-related learning modules.
- Reviewed poster content and revised MongoDB fundamentals, including databases, collections, documents, and data retrieval.
- GitHub Link: https://github.com/JaspinderKaurWalia26/practice_project

Overall, the training enhanced understanding of MongoDB operations, FastAPI backend development, database optimization techniques, AI agent integration, and scalable application development.